SVM-2

Recap - Hard margin Classifier - No errors allowed.

→ Soft Margen Classifier - errors are allowed
penality ?

Hingeloss $\max(0, 1-z_i)$

→ $\underset{\vec{w}, b}{\arg\min} \left[\frac{\|w\|}{2} + C \cdot \frac{1}{N} \sum_{i=1}^{N} \xi_i \right]$ - Primal form ✓

$(\vec{w}, b)$ ✓ Minimum possible

SVM - $\boxed{\text{Kernel Trick}}$ ✓

Primal form → $\boxed{\text{Dual form}}$ ✓

$$\underset{\alpha_i}{\max} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j \, y_i y_j \, x_i^T \cdot x_j$$

α	y (label)	x
weight for each point	-1, $+1$	input vector

$$\left[x_1^T \cdot x_2 \right]$$

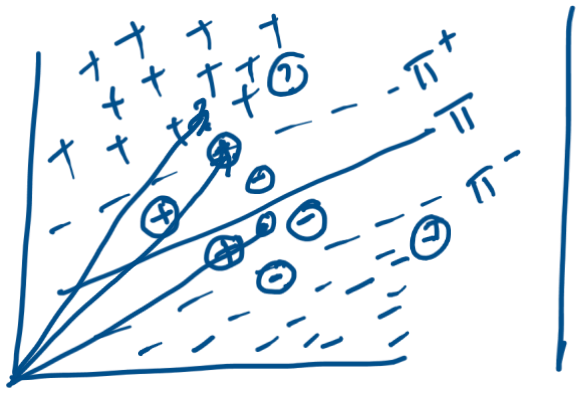
Dot product

$$\cos\theta = \frac{x_1^T \cdot x_2}{\|x_1\| \|x_2\|}$$

<u>Dual form</u> - Simplified represented in the form of Dot products.

$\alpha_i = 0$ for non Support Vectors

$0 < \alpha_i < c$ for all Support vectors.



find out Support vectors.

α_i - for points that are Support vectors.

for Support Vectors $\boxed{\alpha_i > 0}$

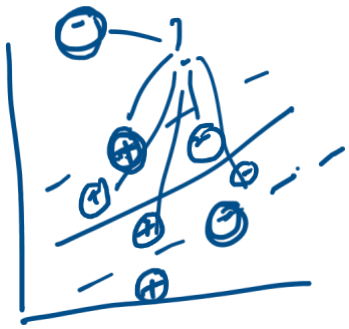
for non Support vectors $\alpha_i = 0$

① Points that fall on the π^+ or π^- hyperplane

② points that fall in the gutter.

③ points that are Misclassified.

→ support vectors? ✓



→ Prediction $\boxed{N = 10{,}000}$ → (100) Support

x_i - all support vectors

$$\underset{f(x_q)}{\hat{y}} : \sum_{i=1}^{N} \underbrace{\alpha_i \, y_i \cdot [x_i^T \cdot x_q]}$$

Training → [Support vectors] →

≥ 0	> 0	0	0
α_1	α_2	$\cancel{\alpha_3}$	$\cdots \cancel{\alpha_{10000}}$
(x_1, y_1)	(x_2, y_2)		

SVM

$$\boxed{\hat{y} = \alpha_1 \cdot y_1 \cdot [x_1^T \cdot x_q] + \alpha_2 \cdot y_2 \cdot [x_2^T \cdot x_q]}$$

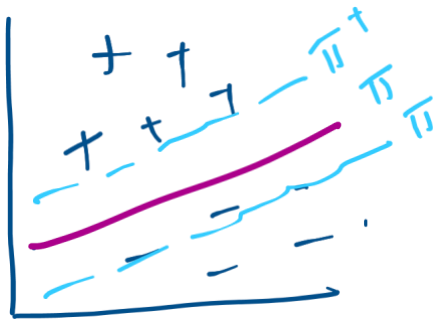
Training → $\boxed{\check{\alpha}_i - [\check{x}_i, \check{y}_i]}$ - key value pair

arrays

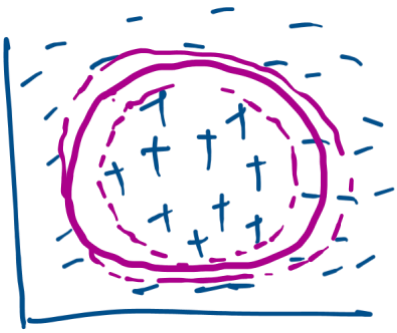
$\check{\alpha}_i \cdot y_i \cdot [x_i^T \cdot x_a]$ ✓ → C++

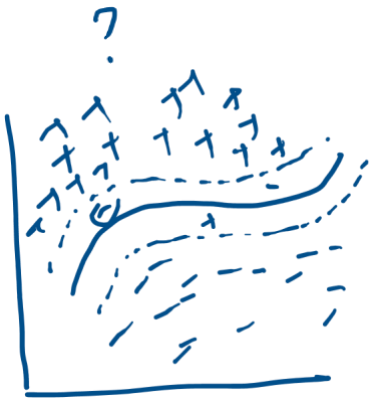
Edge deployments :-

Hyperplane (linear) Seperates this data points



Data (non linear)





→ Why line? Margin as a Classifier ✓

→ It dont allow error Rigid - hinglass - Soft Margin Classifier ✓

Hard Margin

$$\|W\|^2 + C \cdot \frac{1}{N} \sum_{i=1}^{N} \xi_i -$$

→ Primal form — Dual form ? ✓

complex

[x] . [y] . [α] dependent

α_i , [x, y] x_a .

→ Prediction . — learn and remember

α_i [x, y] x_a ? ✓

sup vector

→ Assumption — Polynomial features ?

linear Separation ?

No-linear Separation

[Kernel] SVM ✓

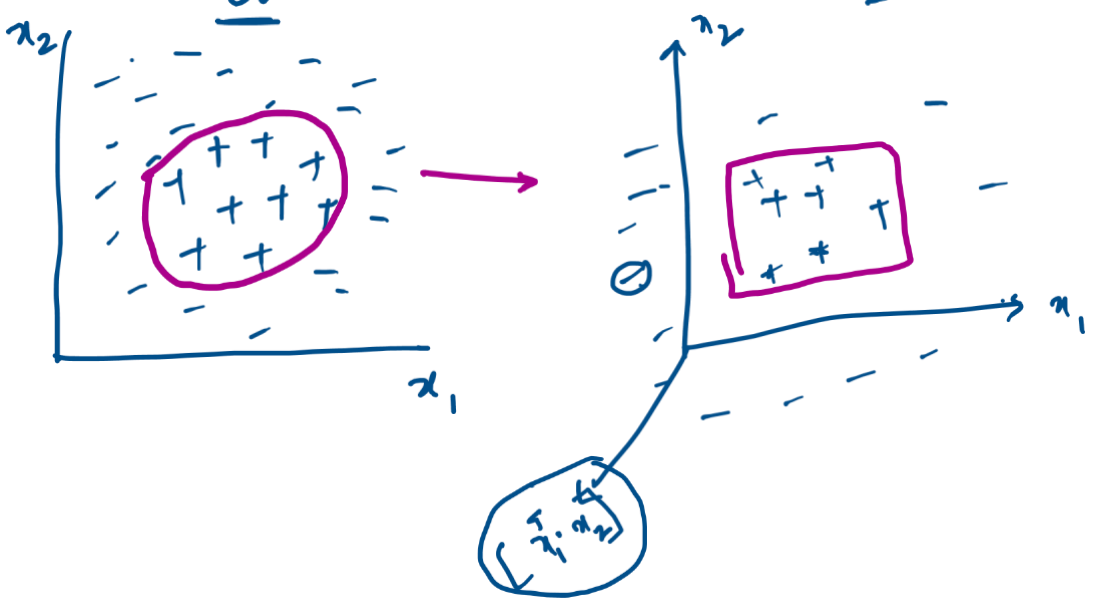
→ Non linear boundary — Polynomial features ✓

Kernels — TRICK

Similarity (x_i, x_j) < Cosine Similarity (x_i, x_j) $[x_i^T \cdot x_j]$

Kernel $[x_i, x_j]$ → Dot products

→ d^2 ⟹ d^3 d' $d' >> d$



① Polynomial Kernel

$(x_1)^2 \quad [x_{11}, x_{12}] \quad x_2 \to [x_{21}, x_{22}]$

$$K(x_1, x_2) = (x_1^T \cdot x_2 + C)^m$$

m-degree

$m = 2$

$m = 2$ - quardratic kernel. , $C = 1$

$$\left(1 + [x_{11} x_{12}] \cdot \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}\right)^2$$

$$(1 + x_{11}x_{21} + x_{12}x_{22})^2 \qquad (a+b+c)^2$$

$$= \left[1 + x_{11}^2 \cdot x_{21}^2 + x_{12}^2 \cdot x_{22}^2 + 2 \cdot x_{11}x_{21} + 2x_{12}x_{22} + 2x_{11}x_{21}x_{12}x_{22}\right] \to \text{Kernel}$$

$\alpha_i \cdot y_i (x_i^T \cdot x_j) \cdot x_a$

LD $\longrightarrow$ K_q $\longrightarrow$ $\overset{b}{\boxed{ND}}$ High dimentional space

$\to$ performed as dot prod

RBF $\to$ Radial Basis function.

linear kernel $\to$ cosine $[x_i^T \cdot x_j]$

$$K_{RBF}(x_1, x_2) = e^{-\frac{\|x_1 - x_2\|^2}{2\sigma^2}}$$

$\leftarrow$ euclidean distance

σ hyperparameter

→ Euclidean distance

$e^{-\text{euclidean dis}}$

$x_1 \longleftrightarrow x_2$

distance ~ 0

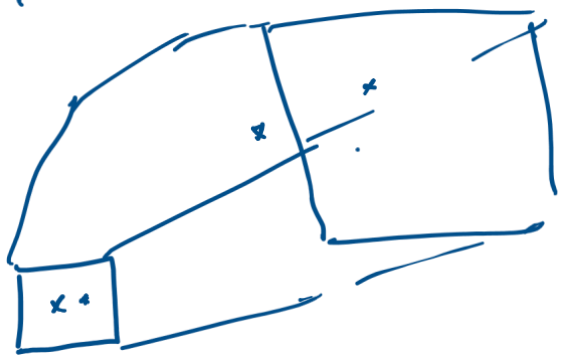
$K_{RBF}(x_1, x_2) \rightarrow$ high

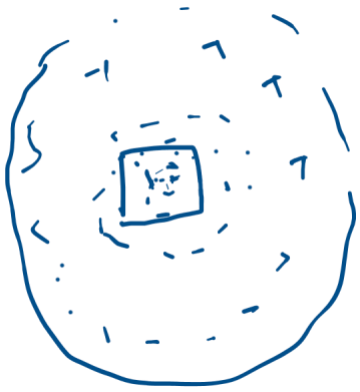
$e^{-0.01} = \frac{1}{e^{0.01}}$

$(x_1 \longleftrightarrow x_3)$

$e^{-1} \quad \frac{1}{2.7} \sim 0.3$

$RBF(x_1 \rightarrow x_3)$ - low





Training

$\sum_{i=1}^{N} \sum_{j=1}^{N} x_i^T \cdot x_j$

$O(n^2)$

slow

Testing

$\alpha_i \cdot y_i \, x_i^T \, x_q$

When $\alpha_i > 0$ - only for sv

$m \times 1$

→ $\boxed{\text{Kernel Tricks}}$ ✓

linear
polynomial
RBF
(sigmoid.)* ✓

ML